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Mathematics-IV (PDE, Probability and Statistics)

BAS303

One-day revision notes: definitions, solved examples, code and important questions

Program: B.Tech | **Branches:** CSE, IT, CSE-AIML, CSE-DS, AIDS, ECE, EE, ME, CHE | **Semester:** 3

Credits: 4 | **L-T-P:** 3-1-0

Scheme: AKTU B.Tech 2nd year, 2023-24 onwards (BAS403 in semester 4 for some branches)

UNIT 1 Partial Differential Equations

UNIT 2 Applications of PDE and Fourier Transform

UNIT 3 Statistical Techniques I

UNIT 4 Statistical Techniques II (Probability)

UNIT 5 Statistical Techniques III (Sampling and SQC)

Original revision notes written for the AKTU syllabus. Maths is about methods, so every unit has:

1. **2-mark definitions and formulas**: what to write in Section A.
2. **Core methods**: step-by-step recipes for each type of question.
3. **Solved examples**: fully worked problems of the kinds asked in AKTU papers.
4. **Formula sheet**: everything to remember for the unit, in one block.
5. **Important questions**: the question types to practise.

How to use these notes the day before the paper

- **Morning (3 hours):** Units 1 and 2. Practise Lagrange's method, one Charpit problem, finding CF and PI, and one heat/wave equation problem.
- **Afternoon (3 hours):** Units 3 and 4: moments, curve fitting, regression, and the binomial/Poisson/normal problems. These are quick, reliable marks.
- **Evening (2 hours):** Unit 5 (t-test, chi-square, control charts), then the formula sheets once more.
- **In the exam:** write the formula first, then substitute. State the hypothesis and the conclusion in words for every test. Keep a table for statistics data.

AKTU paper pattern (theory, 70 marks, 3 hours): Section A has 7 short questions of 2 marks, all compulsory (14). Section B has 5 questions of 7 marks, attempt any 3 (21). Section C has 5 questions (one per unit), each with two parts, attempt one part of each, 7 marks each (35). Every unit gets its own Section C question, so do not skip any unit. (This matches the 2023-24 BAS303 paper.)

UNIT 1 Partial Differential Equations

✍️ 2-mark definitions and formulas

- **PDE:** an equation involving a function of two or more independent variables and its partial derivatives. Notation: $p = \partial z/\partial x$, $q = \partial z/\partial y$, $r = \partial^2 z/\partial x^2$, $s = \partial^2 z/\partial x \partial y$, $t = \partial^2 z/\partial y^2$.
- **Order:** the highest derivative present. **Degree:** the power of the highest-order derivative (after removing radicals and fractions).
- **Linear PDE:** z and its derivatives appear only to the first power and are not multiplied together.
- **Formation of PDE:** eliminate arbitrary constants (differentiate and eliminate) or arbitrary functions.
- **Complete integral:** solution with as many arbitrary constants as independent variables. **General integral:** solution containing an arbitrary function. **Singular integral:** obtained by eliminating constants from the complete integral and its partial derivatives with respect to the constants.
- **Lagrange's linear equation:** $Pp + Qq = R$, where P, Q, R are functions of x, y, z.
- **Classification of A u_{xx} + B u_{xy} + C u_{yy} + ... = 0:** $B^2 - 4AC < 0$ **elliptic**, $= 0$ **parabolic**, > 0 **hyperbolic**.

Core methods

Lagrange's method (Pp + Qq = R)

1. Write the auxiliary equations **dx/P = dy/Q = dz/R**.
2. Find two independent solutions $u(x, y, z) = c_1$ and $v(x, y, z) = c_2$, using grouping or **multipliers** (choose l, m, n with lP + mQ + nR = 0, so l dx + m dy + n dz = 0).
3. General solution: **φ(u, v) = 0**.

Charpit's method (non-linear f(x, y, z, p, q) = 0)

1. Auxiliary equations: **dp/(f_x + p f_z) = dq/(f_y + q f_z) = dz/(−p f_p − q f_q) = dx/(−f_p) = dy/(−f_q)**.
2. Use the simplest pair to get one extra relation between p and q (with a constant a).
3. Solve it with f = 0 for p and q, then integrate **dz = p dx + q dy**.

Standard forms (shortcuts)

Form	Complete integral
I: f(p, q) = 0	z = ax + by + c, where f(a, b) = 0
II (Clairaut): z = px + qy + f(p, q)	z = ax + by + f(a, b)
III: f(z, p, q) = 0	Put q = ap, u = x + ay; solve the ODE in z(u)
IV: $f_1(x, p) = f_2(y, q)$	Set both = a; $p = g_1(x, a)$, $q = g_2(y, a)$; $z = \int p\,dx + \int q\,dy + b$

Higher-order linear PDE with constant coefficients: F(D, D′)z = f(x, y), $D = \partial/\partial x$, $D' = \partial/\partial y$.

- **Complementary function (CF):** put D = m, D′ = 1 to get the auxiliary equation.
- Distinct roots $m_1, m_2, \ldots$: $CF = \varphi_1(y + m_1 x) + \varphi_2(y + m_2 x) + \ldots$

- Repeated root m (twice): CF = $\varphi_1(y + mx) + x\,\varphi_2(y + mx)$.
- **Particular integral (PI) = f(x, y) / F(D, D′):**
- f = e^(ax+by): PI = e^(ax+by) / F(a, b), if F(a, b) ≠ 0.
- f = sin(ax + by) or cos(ax + by): replace $D^2 \to -a^2$, $DD' \to -ab$, $D'^2 \to -b^2$.
- f = $x^m y^n$: expand 1/F(D, D′) as a power series and operate.
- General case: $1/(D - mD')\, f(x, y) = \int f(x, c - mx)\, dx$, then put c = y + mx.
- **Complete solution z = CF + PI.**

Equations reducible to constant coefficients (Euler–Cauchy type): for terms like x^2r, xy s, y^2t, put x = e^X, y = e^Y; then xD = θ, yD′ = θ′, $x^2D^2 = \theta(\theta - 1)$, $xyDD' = \theta\theta'$, $y^2D'^2 = \theta'(\theta' - 1)$.

🧮 Solved examples

Example 1 (formation). Eliminate the arbitrary function from z = f(2x − y). p = 2f′(2x − y), q = −f′(2x − y). Divide: p/q = −2, so **p + 2q = 0**.

Example 2 (Lagrange). Solve (y − z)p + (z − x)q = x − y.

- Multipliers 1, 1, 1: P + Q + R = 0 ⇒ dx + dy + dz = 0 ⇒ $\mathbf{x + y + z = c_1}$.
- Multipliers x, y, z: xP + yQ + zR = 0 ⇒ x dx + y dy + z dz = 0 ⇒ $\mathbf{x^2 + y^2 + z^2 = c_2}$.
- Solution: $\boldsymbol{\varphi(x + y + z, x^2 + y^2 + z^2) = 0}$.

Example 3 (Charpit). Solve px + qy = pq.

- f = px + qy − pq; $f_x = p$, $f_y = q$, $f_z = 0$, $f_p = x - q$, $f_q = y - p$.
- From dp/p = dq/q: **p = aq**.
- Put in f = 0: $aqx + qy - aq^2 = 0 \Rightarrow q = (ax + y)/a$, $p = ax + y$.
- $dz = (ax + y)\,dx + ((ax + y)/a)\,dy = (1/a)(ax + y)(a\,dx + dy)$.
- Integrate: $\mathbf{z = (ax + y)^2/(2a) + b}$.

Example 4 (CF and PI). Solve $(D^2 + 3DD' + 2D'^2)z = x + y$.

- Auxiliary: $m^2 + 3m + 2 = 0 \Rightarrow m = -1, -2$. $\mathbf{CF = \varphi_1(y - x) + \varphi_2(y - 2x)}$.
- $PI = (1/D^2)[1 + 3D'/D + 2D'^2/D^2]^{-1}(x + y) = (1/D^2)[(x + y) - 3(1/D)(1)] = x^3/6 + x^2y/2 - x^3/2$.
- $\mathbf{PI = x^2y/2 - x^3/3}$. (Check: $D^2(PI) = y - 2x$, $DD'(PI) = x$, $D'^2(PI) = 0$; sum = y − 2x + 3x = x + y ✔.)
- $\mathbf{z = \varphi_1(y - x) + \varphi_2(y - 2x) + x^2y/2 - x^3/3}$.

Example 5 (classification). $u_{xx} + 4u_{xy} + 4u_{yy} = 0$: $B^2 - 4AC = 16 - 16 = 0 \Rightarrow$ **parabolic**. $u_{xx} + u_{yy} = 0$: $0 - 4 < 0 \Rightarrow$ **elliptic**. $u_{tt} - c^2u_{xx} = 0$: $0 + 4c^2 > 0 \Rightarrow$ **hyperbolic**.

Formula sheet

```
p = ∂z/∂x, q = ∂z/∂y, r = ∂²z/∂x², s = ∂²z/∂x∂y, t = ∂²z/∂y²
Lagrange:  dx/P = dy/Q = dz/R   →  φ(u, v) = 0
Charpit:   dp/(fx + p fz) = dq/(fy + q fz) = dz/(-p fp - q fq) = dx/(-fp) = dy/(-fq)
CF: distinct m → φ(y + m x);  repeated m → φ1(y + m x) + x φ2(y + m x)
PI: e^(ax+by)/F(a,b);  sin/cos: D² → -a², DD' → -ab, D'² → -b²
Class: B² - 4AC  <0 elliptic, =0 parabolic, >0 hyperbolic
```

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⭐ Important questions

1. Form PDEs by eliminating arbitrary constants / functions.
2. Solve Lagrange's linear equations using grouping and multipliers.
3. Solve non-linear PDEs using Charpit's method or standard forms.
4. Solve higher-order linear PDEs with constant coefficients (CF + PI, including exponential, trigonometric and polynomial right-hand sides).
5. Classify second-order PDEs.

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UNIT 2 Applications of PDE and Fourier Transform

✍️ 2-mark definitions and formulas

- **Method of separation of variables:** assume u(x, t) = X(x)T(t), substitute, and split the PDE into ODEs in each variable equal to a separation constant ($-\lambda^2$).
- **One-dimensional heat equation:** $\partial u/\partial t = c^2\, \partial^2 u/\partial x^2$ (parabolic), c^2 = thermal diffusivity.
- **One-dimensional wave equation:** $\partial^2 y/\partial t^2 = c^2\, \partial^2 y/\partial x^2$ (hyperbolic), $c^2 = T/m$ (tension / mass per unit length).
- **Two-dimensional Laplace equation:** $\partial^2 u/\partial x^2 + \partial^2 u/\partial y^2 = 0$ (steady-state 2D heat flow, elliptic).
- **Fourier transform:** $F\{f(x)\} = F(s) = \int_{-\infty}^{\infty} f(x)\, e^{isx}\, dx$; inverse $f(x) = (1/2\pi) \int_{-\infty}^{\infty} F(s)\, e^{-isx}\, ds$.
- **Fourier sine transform:** $F_s(s) = \int_0^{\infty} f(x) \sin sx\, dx$; inverse $f(x) = (2/\pi) \int_0^{\infty} F_s(s) \sin sx\, ds$.
- **Fourier cosine transform:** $F_c(s) = \int_0^{\infty} f(x) \cos sx\, dx$; inverse $f(x) = (2/\pi) \int_0^{\infty} F_c(s) \cos sx\, ds$.
- **Convolution:** $(f * g)(x) = \int_{-\infty}^{\infty} f(u)\, g(x-u)\, du$. **Convolution theorem:** $F\{f * g\} = F(s) \cdot G(s)$.

Core methods

Choosing the right solution (bounded, physically sensible)

Equation	Solution form to use
Heat ($u \to 0$ as $t \to \infty$)	$u = (A \cos px + B \sin px)\, e^{-c^2 p^2 t}$
Wave (periodic in t)	$y = (A \cos px + B \sin px)(C \cos cpt + D \sin cpt)$
Laplace	$u = (A \cos px + B \sin px)(C e^{py} + D e^{-py})$ or with sinh/cosh

Standard results (rod/string of length L, ends kept at zero)

- **Heat:** $u(x, t) = \Sigma\, b_n \sin(n\pi x/L)\, e^{-c^2 n^2 \pi^2 t/L^2}$, with $\mathbf{b_n = (2/L) \int_0^L f(x) \sin(n\pi x/L)\, dx}$, where $f(x) = u(x, 0)$.
- **Wave:** $y(x, t) = \Sigma\, [a_n \cos(n\pi ct/L) + b_n \sin(n\pi ct/L)] \sin(n\pi x/L)$, with $\mathbf{a_n = (2/L) \int_0^L f(x) \sin(n\pi x/L)\, dx}$ (initial displacement) and $\mathbf{b_n = (2/(n\pi c)) \int_0^L g(x) \sin(n\pi x/L)\, dx}$ (initial velocity). Released from rest ⇒ all $b_n = 0$.
- **Laplace on a rectangle $0 \le x \le a$, $0 \le y \le b$**, with $u = 0$ on $x = 0$, $x = a$, $y = 0$ and $u(x, b) = f(x)$: $u = \Sigma\, B_n \sin(n\pi x/a) \sinh(n\pi y/a)$, $\mathbf{B_n = [2 / (a \sinh(n\pi b/a))] \int_0^a f(x) \sin(n\pi x/a)\, dx}$.

Useful transforms

f(x)	Transform
$e^{-a\lvert x\rvert}$	$F(s) = 2a/(a^2 + s^2)$
e^{-ax}, $x > 0$	$F_s = s/(a^2 + s^2)$, $F_c = a/(a^2 + s^2)$
1 for $\lvert x\rvert < a$, 0 otherwise	$F(s) = 2 \sin(as)/s$
$e^{-x^2/2}$	$F(s) = \sqrt{(2\pi)}\, e^{-s^2/2}$

Transform of derivatives (for solving PDEs): $F\{\partial^2u/\partial x^2\} = -s^2U$; $F_s\{u_{xx}\} = s\,u(0) - s^2U_s$; $F_c\{u_{xx}\} = -u_x(0) - s^2U_c$. Pick sine transform when u(0, t) is given, cosine transform when $u_x(0, t)$ is given.

🧮 Solved examples

Example 1 (wave equation). A string fixed at x = 0 and x = 2 is released from rest from y(x, 0) = a sin(πx/2). Find y(x, t).

- L = 2; at rest ⇒ $b_n = 0$. The initial shape is already a single term with n = 1, so $a_1 = a$ and all other $a_n = 0$.
- **y(x, t) = a sin(πx/2) cos(πct/2)**.

Example 2 (heat equation). $u_t = c^2u_{xx}$, $0 < x < \pi$, $u(0, t) = u(\pi, t) = 0$, $u(x, 0) = 3 \sin x + 5 \sin 4x$.

- L = π, so $u = \Sigma\, b_n \sin nx\, e^{(-c^2n^2t)}$. Matching the initial condition directly: $b_1 = 3$, $b_4 = 5$.
- **u(x, t) = 3 sin x e^(−c²t) + 5 sin 4x e^(−16c²t)**.

Example 3 (separation of variables on a first-order PDE). Solve $u_x = 2u_t + u$, $u(x, 0) = 6e^{(-3x)}$.

- $u = X(x)T(t)$: $X'T = 2XT' + XT \Rightarrow X'/X = (2T' + T)/T = k$.
- $X = Ae^{(kx)}$; $2T' = (k - 1)T \Rightarrow T = Be^{((k-1)t/2)}$. $u = C\, e^{(kx)}\, e^{((k-1)t/2)}$.
- At t = 0: $Ce^{(kx)} = 6e^{(-3x)} \Rightarrow C = 6$, $k = -3$. **u = 6e^(−3x − 2t)**.

Example 4 (Fourier transform). Find $F\{e^{(-a|x|)}\}$, $a > 0$. $F(s) = \int_{-\infty}^{0} e^{(ax)}\, e^{(isx)}\, dx + \int_0^{\infty} e^{(-ax)}\, e^{(isx)}\, dx = 1/(a + is) + 1/(a - is) = \mathbf{2a/(a^2 + s^2)}$.

Formula sheet

```
Heat:    u = Σ bn sin(nπx/L) e^(-c²n²π²t/L²),       bn = (2/L)∫ f sin(nπx/L) dx
Wave:    y = Σ [an cos(nπct/L) + bn sin(nπct/L)] sin(nπx/L)
         an = (2/L)∫ f sin(nπx/L) dx,   bn = (2/(nπc))∫ g sin(nπx/L) dx
Laplace: u = Σ Bn sin(nπx/a) sinh(nπy/a)
FT:  F(s) = ∫ f e^(isx) dx;  inverse (1/2π)∫ F e^(-isx) ds
FST/FCT inverse factor 2/π;   F{f*g} = F·G;   F{u_xx} = -s²U
```

⭐ Important questions

1. Solve PDEs by the method of separation of variables.
2. Solve the one-dimensional heat equation with given boundary and initial conditions.
3. Solve the one-dimensional wave equation for a vibrating string (released from rest / given initial velocity).
4. Solve Laplace's equation in a rectangle.
5. Find Fourier, sine and cosine transforms of given functions; state and prove the convolution theorem; solve a PDE using Fourier transforms.

UNIT 3 Statistical Techniques I

✍️ 2-mark definitions and formulas

- **Measures of central tendency:** mean $\bar{x} = \Sigma fx/N$, median (middle value), mode (most frequent). Empirical relation: **Mode = 3 Median − 2 Mean**.
- **Moments about the mean:** $\mu_r = \Sigma f(x - \bar{x})^r/N$. **Moments about A:** $\mu_r' = \Sigma f(x - A)^r/N$.
- **Skewness:** lack of symmetry. Karl Pearson's coefficient S_k = (Mean − Mode)/σ. Moment measure $\boldsymbol{\beta_1 = \mu_3^2/\mu_2^3}$; sign of μ_3 gives the direction.
- **Kurtosis:** peakedness. $\boldsymbol{\beta_2 = \mu_4/\mu_2^2}$: = 3 mesokurtic (normal), > 3 leptokurtic, < 3 platykurtic.
- **Curve fitting (least squares):** choose constants so that the sum of squares of errors $\Sigma(y - \hat{y})^2$ is minimum.
- **Correlation coefficient (Karl Pearson):** $r = \mathrm{Cov}(x, y)/(\sigma_x\sigma_y)$, $-1 \le r \le 1$.
- **Rank correlation (Spearman):** $\rho = 1 - 6\Sigma d^2/(n(n^2 - 1))$.
- **Regression line:** the best-fit line to predict one variable from the other. Regression coefficients $b_{yx} = r\,\sigma_y/\sigma_x$ and $b_{xy} = r\,\sigma_x/\sigma_y$; $\boldsymbol{r = \pm\sqrt{(b_{yx}\,b_{xy})}}$.

Core methods

Moments about the mean from moments about A

- $\mu_1 = 0$
- $\mu_2 = \mu_2' - \mu_1'^2$
- $\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2\mu_1'^3$
- $\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'\mu_1'^2 - 3\mu_1'^4$
- Mean = $A + \mu_1'$; variance = μ_2.

Normal equations (least squares)

Curve	Normal equations
Line $y = a + bx$	$\Sigma y = na + b\Sigma x$; $\Sigma xy = a\Sigma x + b\Sigma x^2$
Parabola $y = a + bx + cx^2$	$\Sigma y = na + b\Sigma x + c\Sigma x^2$; $\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$; $\Sigma x^2y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$
$y = ae^{bx}$	Take $Y = \ln y$: $Y = A + bx$ with $A = \ln a$, then fit a line
$y = ax^b$	Take logs: $\log y = \log a + b \log x$, fit a line

Regression lines: y on x: $y - \bar{y} = b_{yx}(x - \bar{x})$; x on y: $x - \bar{x} = b_{xy}(y - \bar{y})$. Both pass through $(\bar{x}, \bar{y})$, so the **means are the intersection point**. The angle between them shrinks as $|r| \to 1$.

🧮 Solved examples

Example 1 (moments, skewness, kurtosis). The first four moments about 5 are 2, 20, 40 and 50.

- $\mu_2 = 20 - 2^2 = \mathbf{16}$.

- $\mu_3 = 40 - 3(20)(2) + 2(2)^3 = 40 - 120 + 16 = \mathbf{-64}$.
- $\mu_4 = 50 - 4(40)(2) + 6(20)(4) - 3(16) = 50 - 320 + 480 - 48 = \mathbf{162}$.
- Mean = 5 + 2 = 7. $\beta_1 = (-64)^2/16^3 = 4096/4096 = \mathbf{1}$; $\beta_2 = 162/256 = \mathbf{0.63}$.
- $\mu_3 < 0 \Rightarrow$ **negatively skewed**; $\beta_2 < 3 \Rightarrow$ **platykurtic**.

Example 2 (fit a straight line). x: 1, 2, 3, 4, 5; y: 2, 4, 5, 4, 5.

- n = 5, $\Sigma x = 15$, $\Sigma y = 20$, $\Sigma x^2 = 55$, $\Sigma xy = 66$.
- $20 = 5a + 15b$ and $66 = 15a + 55b \Rightarrow b = 0.6$, $a = 2.2$.
- $\mathbf{y = 2.2 + 0.6x}$.

Example 3 (regression lines). The lines are $8x - 10y + 66 = 0$ and $40x - 18y = 214$, and Var(x) = 9. Find $\bar{x}$, $\bar{y}$, r and σ_y.

- Solving together: $\mathbf{\bar{x} = 13, \bar{y} = 17}$.
- Take the first as y on x: $y = 0.8x + 6.6 \Rightarrow b_{yx} = 0.8$. Second as x on y: $x = 0.45y + 5.35 \Rightarrow b_{xy} = 0.45$.
- $r^2 = 0.8 \times 0.45 = 0.36 \Rightarrow \mathbf{r = 0.6}$ (positive, since both coefficients are positive; and $r^2 \le 1$ confirms the choice).
- $\sigma_y = b_{yx}\,\sigma_x / r = 0.8 \times 3 / 0.6 = \mathbf{4}$.

Example 4 (rank correlation). For 5 pairs with $\Sigma d^2 = 8$: $\rho = 1 - 6(8)/(5 \times 24) = 1 - 0.4 = \mathbf{0.6}$.

Formula sheet

```
μ2 = μ2' - μ1'²;  μ3 = μ3' - 3μ2'μ1' + 2μ1'³;  μ4 = μ4' - 4μ3'μ1' + 6μ2'μ1'² - 3μ1'⁴
β1 = μ3²/μ2³;  β2 = μ4/μ2²  (3 = normal)
r = Σ(dx·dy)/√(Σdx²·Σdy²);   ρ = 1 - 6Σd²/(n(n²-1))
byx = r σy/σx;  bxy = r σx/σy;  r = ±√(byx·bxy)
Line normal eqns: Σy = na + bΣx;  Σxy = aΣx + bΣx²
```

⭐ Important questions

1. Calculate moments, skewness and kurtosis from given data or from moments about a point.
2. Fit a straight line / second-degree parabola / exponential curve by least squares.
3. Find Karl Pearson's correlation coefficient and Spearman's rank correlation.
4. Find regression lines; or from given regression lines find means, r and standard deviations.

UNIT 4 Statistical Techniques II (Probability and Distributions)

✍️ 2-mark definitions and formulas

- **Probability:** P(A) = favourable outcomes / total equally likely outcomes; $0 \le P(A) \le 1$. Addition: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. Conditional: $P(A|B) = P(A \cap B)/P(B)$.
- **Random variable:** a function that assigns a number to each outcome. **Discrete** (countable values) or **continuous** (any value in an interval).
- **Probability mass function (PMF):** $p(x) \ge 0$ and $\Sigma p(x) = 1$.
- **Probability density function (PDF):** $f(x) \ge 0$ and $\int f(x)\,dx = 1$; $P(a < X < b) = \int_a^b f(x)\,dx$.
- **Expectation:** $E(X) = \Sigma x\,p(x)$ or $\int x\,f(x)\,dx$. **Variance:** $Var(X) = E(X^2) - [E(X)]^2$.
- **Binomial distribution:** $P(X = r) = {}^nC_r\,p^r\,q^{n-r}$, mean np, variance npq.
- **Poisson distribution:** $P(X = r) = e^{-\lambda}\,\lambda^r / r!$, mean = variance = λ. Limit of binomial when n is large and p small (λ = np).
- **Normal distribution:** bell-shaped, symmetric about μ; standard variate **$z = (x - \mu)/\sigma$**; mean = median = mode.

Core methods

Choosing the distribution

Situation	Use
Fixed number n of independent trials, two outcomes	Binomial
Rare events in a large number of trials / per unit time	Poisson
Continuous measurements (heights, wages, marks)	Normal

Normal-distribution procedure: convert to z, read the area from 0 to z in the table, then add or subtract from 0.5. Key areas: z = 1 → 0.3413, z = 1.96 → 0.4750, z = 2 → 0.4772, z = 3 → 0.49865.

Fitting distributions: binomial: find p from mean = np, then expected frequencies $N \cdot P(X = r)$. Poisson: λ = mean, then $N \cdot e^{-\lambda}\lambda^r/r!$.

🧮 Solved examples

Example 1 (PDF constant). $f(x) = kx$ for $0 \le x \le 4$. $\int_0^4 kx\,dx = 8k = 1 \Rightarrow$ **$k = 1/8$**. Mean $= \int_0^4 x \cdot (x/8)\,dx = 64/24 =$ **8/3**.

Example 2 (binomial). 320 families have 5 children each; boys and girls are equally likely.

- $P(2 \text{ boys, } 3 \text{ girls}) = {}^5C_2\,(1/2)^5 = 10/32 \Rightarrow$ expected families $= 320 \times 10/32 =$ **100**.
- $P(\text{at least one boy}) = 1 - (1/2)^5 = 31/32 \Rightarrow 320 \times 31/32 =$ **310**.

Example 3 (Poisson). If $P(X = 2) = 9P(X = 4) + 90P(X = 6)$, find the standard deviation.

- $e^{-m}m^2/2 = 9e^{-m}m^4/24 + 90e^{-m}m^6/720 \Rightarrow 1/2 = 3m^2/8 + m^4/8$.

- $m^4 + 3m^2 - 4 = 0 \Rightarrow (m^2 + 4)(m^2 - 1) = 0 \Rightarrow m = 1$.
- Variance = m = 1, so **SD = 1**.

Example 4 (normal). Wages of 1000 workers are normal with mean ₹140 and SD ₹10.

- $140 < x < 144$: z from 0 to 0.4 → area 0.1554 → **≈ 155 workers**.
- $x < 126$: z = −1.4 → area 0.5 − 0.4192 = 0.0808 → **≈ 81 workers**.
- $x > 160$: z = 2 → area 0.5 − 0.4772 = 0.0228 → **≈ 23 workers**.

Example 5 (discrete expectation). X takes 8, 12, 16, 20, 24 with probabilities 1/8, 1/6, 3/8, 1/4, 1/12. E(X) = 1 + 2 + 6 + 5 + 2 = **16**.

Formula sheet

```
Binomial: P(r) = nCr p^r q^(n-r);  mean np;  var npq
Poisson:  P(r) = e^(-λ) λ^r / r!;  mean = var = λ;  recurrence P(r+1) = λ/(r+1)·P(r)
Normal:   z = (x - μ)/σ;  areas: 1→0.3413, 1.96→0.475, 2→0.4772, 3→0.49865
E(X) = Σx p(x);  Var(X) = E(X²) - [E(X)]²
```

⭐ Important questions

1. Find the constant, mean and variance for a given PMF/PDF.
2. Solve problems and fit data using the binomial distribution.
3. Solve problems and fit data using the Poisson distribution.
4. Solve problems on the normal distribution using the area table.

UNIT 5 Statistical Techniques III (Sampling, Tests and SQC)

✍️ 2-mark definitions and formulas

- **Population / sample:** the whole group / a part of it chosen for study. **Parameter:** a value for the population (μ, σ); **statistic:** a value from the sample ($\bar{x}$, s).
- **Sampling distribution / standard error (SE):** the distribution of a statistic over all samples; its SD is the standard error, e.g. $SE(\bar{x}) = \sigma/\sqrt{n}$.
- **Null hypothesis H_o:** a statement of "no difference" that we test. **Alternative hypothesis H_1:** what we accept if H_o is rejected.
- **Type I error:** rejecting a true H_o (probability α, the **level of significance**). **Type II error:** accepting a false H_o (probability β).
- **Critical region:** values of the test statistic for which H_o is rejected. **One-tailed / two-tailed test** depending on H_1.
- **Confidence limits:** $\bar{x} \pm z \cdot SE$; 95% limits use z = 1.96, 99% use 2.58.
- **Degrees of freedom:** number of values free to vary (n − 1 for one-sample t).
- **Statistical quality control (SQC):** using statistics to monitor a process. **Control chart:** a chart with a central line (CL), upper control limit (UCL) and lower control limit (LCL). Points outside the limits mean the process is **out of control** (assignable causes).

Core methods

Steps for any test

1. State H_o and H_1. 2. Choose α (usually 5%). 3. Compute the test statistic. 4. Compare with the table (critical) value. 5. **If |calculated| < table value, accept H_o; otherwise reject.** State the conclusion in words.

Critical z values

α	Two-tailed	One-tailed
5%	1.96	1.645
1%	2.58	2.33

Test statistics

Test	Statistic
Large sample, single mean	$z = (\bar{x} - \mu)/(\sigma/\sqrt{n})$
Large sample, two means	$z = (\bar{x}_1 - \bar{x}_2)/\sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)}$
Single proportion	$z = (p - P)/\sqrt{(PQ/n)}$
Small sample, single mean	$t = (\bar{x} - \mu)/(s/\sqrt{n})$, $s^2 = \Sigma(x - \bar{x})^2/(n - 1)$, df = n − 1

Test	Statistic
Small sample, two means	$t = (\bar{x}_1 - \bar{x}_2)/(S\sqrt{(1/n_1 + 1/n_2)})$, $S^2 = [\Sigma(x_1 - \bar{x}_1)^2 + \Sigma(x_2 - \bar{x}_2)^2]/(n_1 + n_2 - 2)$, $df = n_1 + n_2 - 2$
Chi-square (goodness of fit / independence)	$\chi^2 = \Sigma(O - E)^2/E$; for an r × c table $df = (r - 1)(c - 1)$, E = (row total × column total)/N
2 × 2 table shortcut	$\chi^2 = N(ad - bc)^2 / [(a + b)(c + d)(a + c)(b + d)]$

Critical χ^2 at 5%: 1 df → 3.841, 2 df → 5.991, 3 df → 7.815.

Control charts

Chart	For	CL	Limits
$\bar{X}$ chart	Mean of a variable	$\bar{\bar{X}}$	$\bar{\bar{X}} \pm A_2\bar{R}$
R chart	Range of a variable	$\bar{R}$	UCL = $D_4\bar{R}$, LCL = $D_3\bar{R}$
p chart	Fraction defective	$\bar{p}$	$\bar{p} \pm 3\sqrt{(\bar{p}(1 - \bar{p})/n)}$
np chart	Number defective	$n\bar{p}$	$n\bar{p} \pm 3\sqrt{(n\bar{p}(1 - \bar{p}))}$
c chart	Defects per unit	$\bar{c}$	$\bar{c} \pm 3\sqrt{\bar{c}}$

For sample size n = 5: $A_2 = 0.577$, $D_3 = 0$, $D_4 = 2.114$. A negative LCL is taken as 0.

🧮 Solved examples

Example 1 (t-test, single mean). A sample of 10 has mean 48 and s = 3. Test whether the population mean could be 50 (5% level, $t_{0.05}$ for 9 df = 2.262).

- H_0: $\mu = 50$. $t = (48 - 50)/(3/\sqrt{10}) = -2/0.949 =$ **−2.11**.
- $|t| = 2.11 < 2.262 \Rightarrow$ **accept H_0**: the difference is not significant.

Example 2 (chi-square, independence). 50 workers: men doing skilled work 20, unskilled 10; women skilled 5, unskilled 15. Is the nature of work independent of gender?

- a = 20, b = 10, c = 5, d = 15, N = 50; ad − bc = 300 − 50 = 250.
- $\chi^2 = 50 \times 250^2 / (30 \times 20 \times 25 \times 25) = 3{,}125{,}000 / 375{,}000 =$ **8.33**.
- df = 1, table value 3.841. 8.33 > 3.841 ⇒ **reject H_0**: work type depends on gender.

Example 3 (np chart). 10 samples of 100 items each contain 80 defectives in total.

- $\bar{p} = 80/1000 = 0.08$, $n\bar{p} = 8$, $\sqrt{(n\bar{p}(1 - \bar{p}))} = \sqrt{7.36} = 2.71$.
- **CL = 8, UCL = 8 + 3(2.71) = 16.14, LCL = 8 − 8.14 < 0 → 0.** Any sample with more than 16 defectives signals the process is out of control.

Example 4 ($\bar{X}$ and R charts). Samples of size 5 give $\bar{\bar{X}} = 20$ and $\bar{R} = 4$.

- $\bar{X}$ chart: CL = 20, **UCL = 20 + 0.577 × 4 = 22.31, LCL = 17.69**.
- R chart: CL = 4, **UCL = 2.114 × 4 = 8.46, LCL = 0**.

Formula sheet

```
z (mean) = (x̄ - μ)/(σ/√n);   z (two means) = (x̄1 - x̄2)/√(σ1²/n1 + σ2²/n2)
t = (x̄ - μ)/(s/√n), df n-1;  pooled t, df n1+n2-2
x² = Σ(O-E)²/E;  2×2: N(ad-bc)²/[(a+b)(c+d)(a+c)(b+d)];  df (r-1)(c-1)
z critical: 1.96 (5%), 2.58 (1%) two-tailed
X̄: X̿ ± A2 R̄;  R: D3 R̄, D4 R̄;  p: p̄ ± 3√(p̄q̄/n);  np: np̄ ± 3√(np̄q̄);  c: c̄ ± 3√c̄
```

⭐ Important questions

1. Explain null hypothesis, level of significance, Type I/II errors and critical region.
2. Test the significance of a mean or difference of means using z-test (large samples) or t-test (small samples).
3. Apply the chi-square test for goodness of fit or independence of attributes.
4. Construct $\bar{X}$ and R charts, or p / np / c charts, and comment on whether the process is in control.

Last-hour checklist

Read only this in the last hour before the paper.

- ☐ p, q, r, s, t notation; formation by elimination.
- ☐ Lagrange: dx/P = dy/Q = dz/R; multipliers; φ(u, v) = 0.
- ☐ Charpit auxiliary equations; standard forms I–IV.
- ☐ CF rules (distinct/repeated m); PI for e^(ax+by), sin/cos, polynomials.
- ☐ $B^2 - 4AC$ classification.
- ☐ Heat, wave and Laplace solution formulas with b_n.
- ☐ Fourier, sine, cosine transforms and inverses; convolution theorem.
- ☐ Moment conversion formulas; β_1, β_2.
- ☐ Normal equations for line and parabola.
- ☐ r, ρ, regression coefficients, means from intersection.
- ☐ Binomial, Poisson, normal formulas and table values.
- ☐ Test procedure; z, t, χ^2 formulas; 1.96 / 2.58 / 3.841.
- ☐ Control chart limits; $A_2 = 0.577$, $D_4 = 2.114$ for n = 5.

Books prescribed by AKTU

These are the books named in the AKTU syllabus. Buy or borrow them through a bookshop or your college library. The WorldCat links show which libraries hold a copy.

#	Title	Author(s)	Publisher	Where to get it legally
1	Higher Engineering Mathematics (text book)	B. S. Grewal	Khanna Publishers	WorldCat
2	Advanced Engineering Mathematics	Peter V. O'Neil	Cengage Learning	WorldCat
3	Higher Engineering Mathematics	B. V. Ramana	McGraw-Hill	WorldCat
4	Introductory Methods of Numerical Analysis	S. S. Sastry	PHI Learning	WorldCat
5	Advanced Engineering Mathematics	Erwin Kreyszig	John Wiley & Sons	WorldCat
6	Numerical Methods for Scientific and Engineering Computation	M. K. Jain, S. R. K. Iyengar, R. K. Jain	New Age International	WorldCat
7	An Introduction to Numerical Methods and Analysis	James F. Epperson	Wiley	WorldCat

Official syllabus: AKTU Mathematics-III, IV and V syllabus, 2023-24 scheme (PDF)

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